GFN1000
In Dialogue with Nature

Rachel Louise Carson

Silent spring

Core Question:

What impacts does manipulation of life bring?

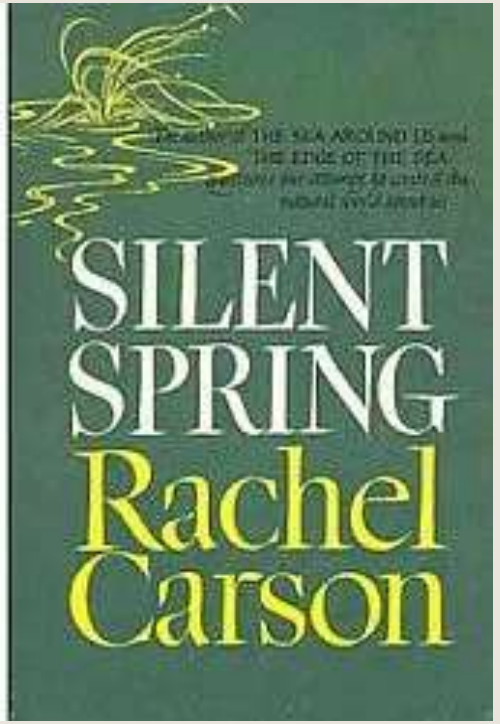




Text:

- Rachel Carson, *Silent Spring*
  - Ch. 6. Earth's green mantle





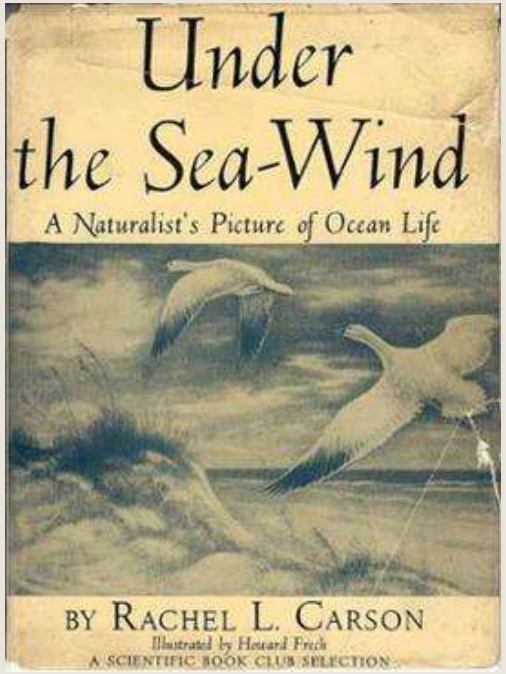
Rachel Louise Carson

- Writer, scientist, and ecologist
- American, born in Pennsylvania
- Bachelor: English, switched to biology in 1928 in Pennsylvania College for Women
- Master: Zoology in Johns Hopkins University (1929 to 1932)
  - Thesis: "*The Development of the Pronephros (前肾) During the Embryonic and Early Larval Life of the Catfish (鲶鱼).*"
- Joined U.S. Bureau of Fisheries (渔业管理局) as a junior aquatic biologist in 1936
  - A fifteen-year career in the federal service as a scientist and editor
  - Later rose to become Editor-in-Chief of all publications for the U. S. Fish and Wildlife Service
  - Became full time nature writer (resigned in 1952)
- *Silent spring* (1962)
- Died of breast cancer in 1964

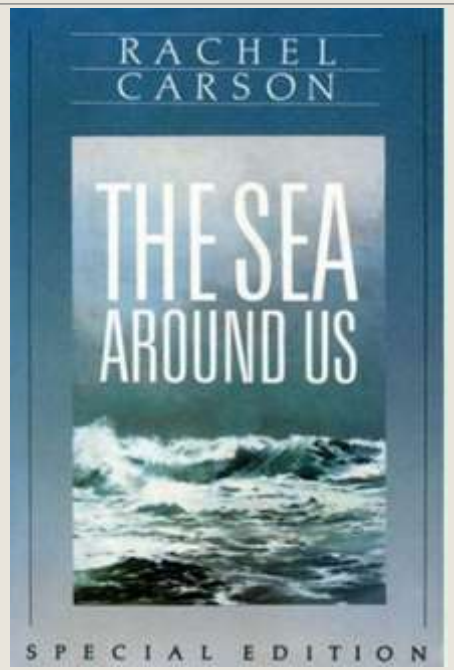


1907-1964

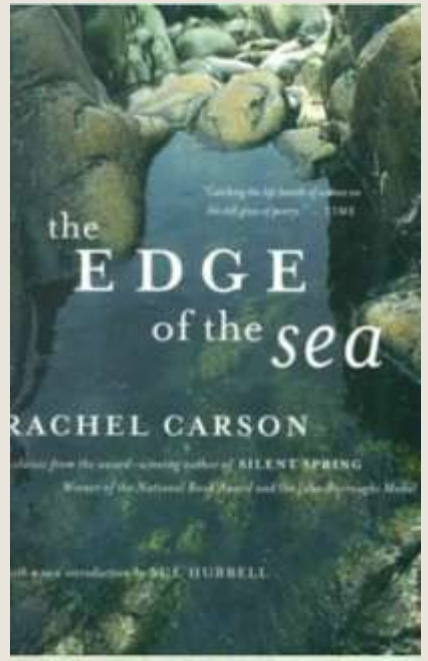
Publications



Under the Sea Wind (1941) 海风下
Narrative of the behavior of organisms that live both on and in the sea on the Atlantic coast.



The Sea around Us (1951)
Life history of the ocean
National Book Award 1952



The Edge of the Sea (1955)
Coastal ecosystem

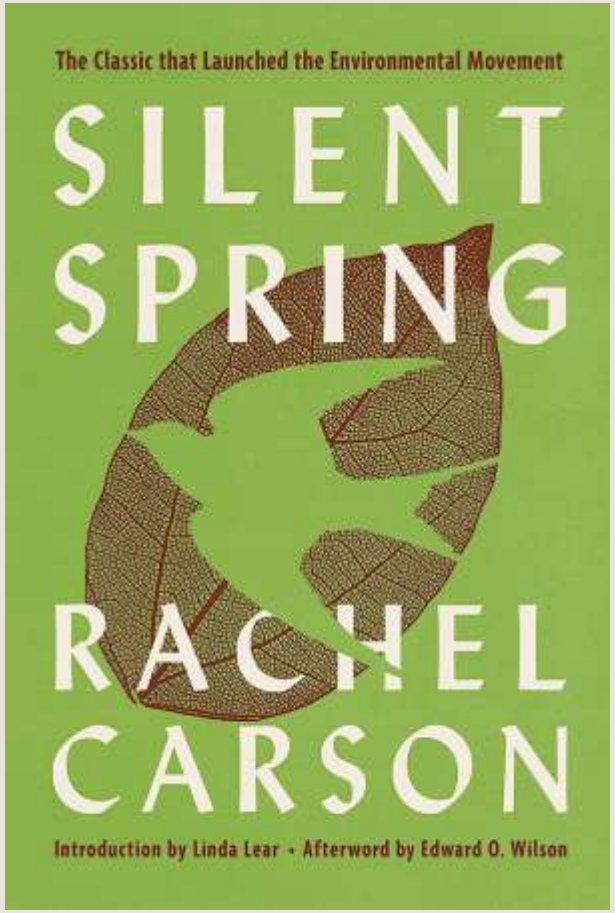
What to write next?

- Contract with publisher – book on evolution
- The turning point: Olga Owens Huckins
  - *"In a letter written in January 1958, Olga Owens Huckins told me of her own bitter experience of a small world made lifeless, and so brought my attention sharply back to a problem with which I had long been concerned."*
- *Silent Spring (1962)*

Silent Spring

Her sensational book *Silent Spring* (1962)

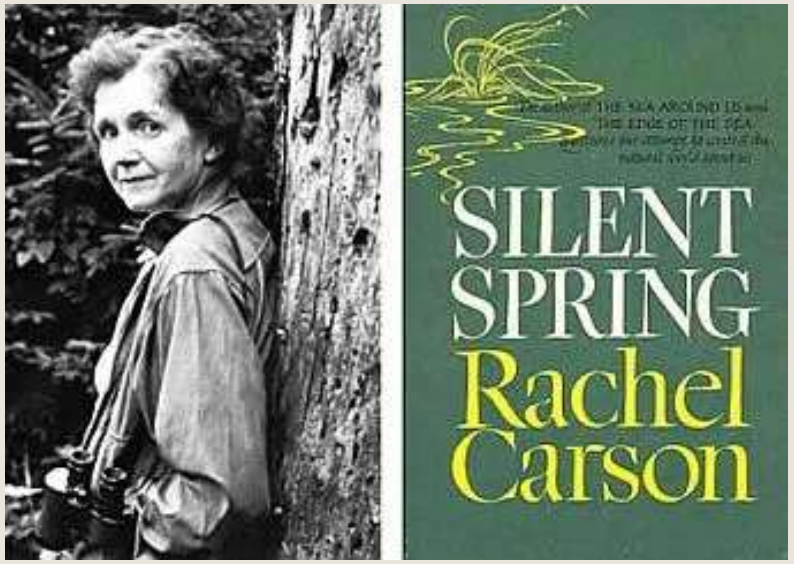
1. A Fable for Tomorrow 明天的寓言
2. The Obligation to Endure? 忍耐的义务
3. Elixirs of Death 死神的特效药
4. Surface Waters and Underground Seas 地表水和地下水
5. Realms of the Soil 土壤的王国
6. **Earth's Green Mantle 地球的绿色斗篷**
7. Needless Havoc 不必要的大破坏
8. And No Birds Sing 再也没有鸟儿歌唱
9. Rivers of Death 死亡的河流
10. Indiscriminately from the Skies 自天而降的灾难
11. Beyond the Dreams of the Borgias 超过了波尔基亚家族的梦想
12. The Human Price 人类的代价
13. Through a Narrow Window 通过一扇狭小的窗户
14. One in Every Four 每四个中有一个
15. Nature Fights Back 大自然在反抗
16. The Rumblings of an Avalanche 崩溃声隆隆
17. The Other Road 另外的道路



Silent Spring

- Selected in:
  - Book-of-the-Month Club
  - Modern Library List of Best 20th-Century Nonfiction (#5)
  - Discover Magazine: 25 Greatest Science Books of All Time
  - The American Reader: Words that Moved a Nation

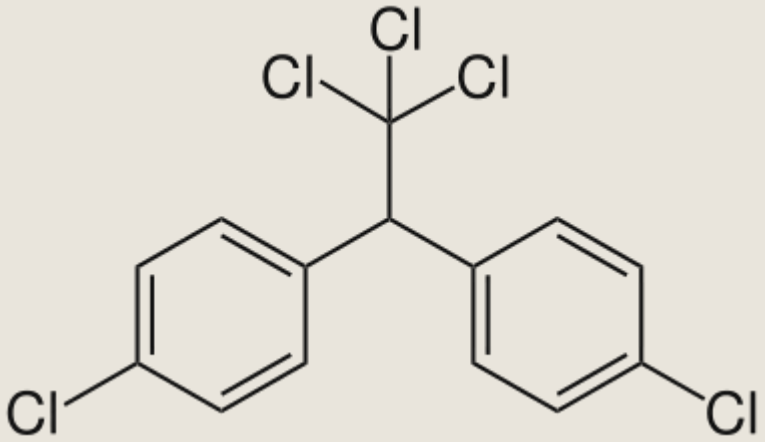
- **Landmark** in the modern environmental movement
  - Ignite the public heightened debate and support toward a new environmentalism



Pesticides

DDT as a pesticide

- **D**ichloro-**D**iphenyl-**T**richloro-ethane (双对氯苯基三氯乙烷)
- Colorless, crystalline solid (无色晶体)
- First synthesized in 1874 by Austrian chemist Othmar Zeidler



DDT

- In 1939, Swiss chemist Paul Müller discovered DDT can be used as an insecticide.
- Effective against a wide variety of pests, including the common house fly, louse (虱子), Colorado beetle, and mosquito. During World War II: successfully controlled typhus (斑疹伤寒) and malaria (疟疾) in Europe
- Agricultural use of DDT since 1945
- Nobel Prize in Physiology or Medicine in 1948
- Worldwide malaria (疟疾) eradication program by WHO in 1955



Paul Müller
(1899-1965)

General Beliefs about Pesticides

- Kills pests only
- Harmless to humans
- Protects crops and controls disease-carrying pests

- A solution to multiple problems

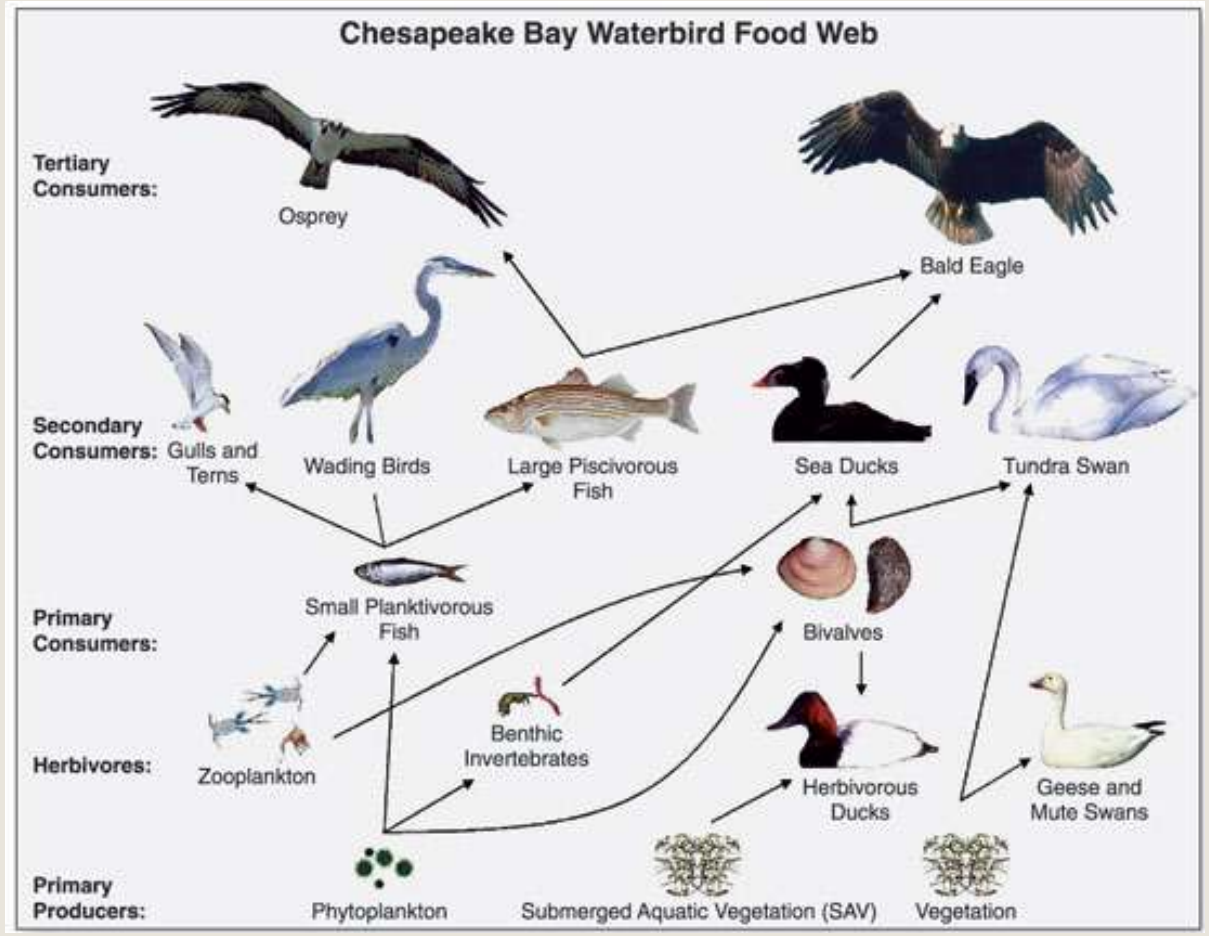
Bioaccumulation (生物累积)

Bioaccumulation is the gradual accumulation of substances, such as pesticides or other chemicals, in an organism.

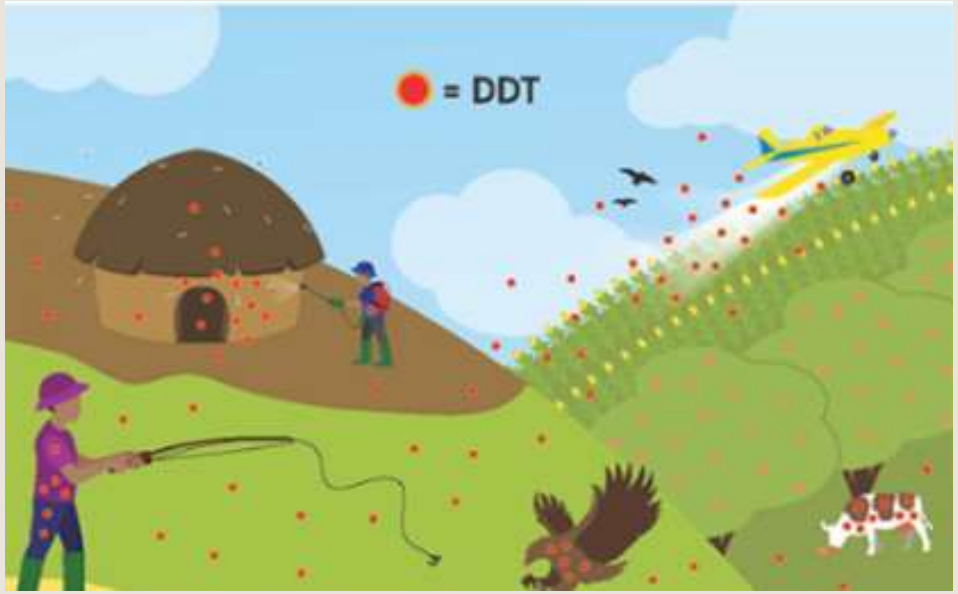


Biomagnification (生物富集)

--concentration of poison along the food chain to lethal levels



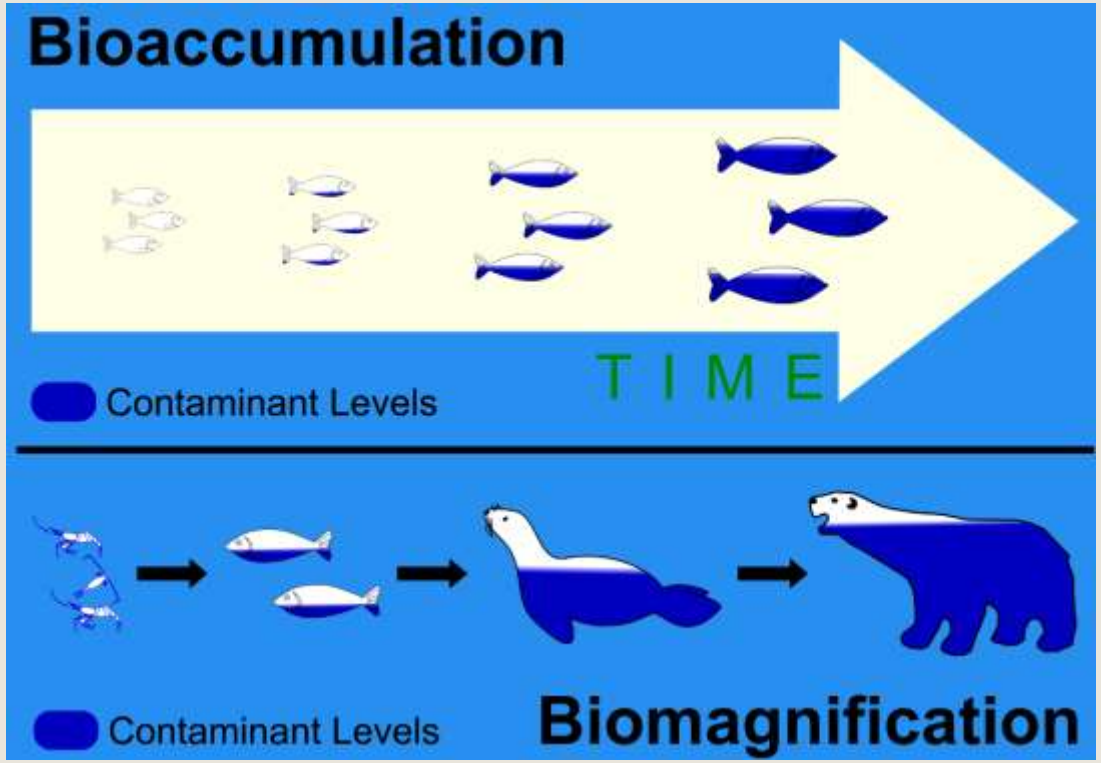
A case in California



	Pesticide Concentration (ppm)
Water	1/50
Plankton	5
Herbivore fish	40 – 300
Carnivore bird and fish	1600 and 2500 respectively

1 ppm = 1 mg/Liter.

Difference of bioaccumulation and biomagnification



Health Hazards

- Persistent Organic Pollutants (POPs)
  - They are resistant to environmental degradation through chemical, biological, and photolytic processes.
- Direct poisoning
  - In animal tests:
    - 3 ppm of DDT inhibits heart muscle enzyme (心肌酶)
    - 5 ppm of DDT causes liver cell necrosis (坏疽)
  - In humans:
    - Nerve system damage (23 ppm of DDT)
- Carcinogenicity (致癌性)
- Genetic damage
- Resistance
  - Natural selection
    - Pesticides: eliminate the members that couldn’t resist pesticides.

1 ppm = 1 mg/Liter.

How about herbicides?

understand these things, but these are not the men who order the wholesale drenching of the landscape with chemicals.

And where are the men who supposedly understand the value of proper habitat for the preservation of wildlife? Too many of them are to be found defending herbicides as "harmless" to wildlife because they are thought to be less toxic than insecticides. Therefore, it is said, no harm is done. But as the herbicides rain down on forest and field, on marsh and rangeland, they are bringing about marked changes and even permanent destruction of wildlife habitat. To destroy the homes and the food of wildlife is perhaps worse in the long run than direct killing.

The irony of this all-out chemical assault on roadsides and utility rights-of-way is twofold. It is perpetuating the problem it seeks to correct, for as experience

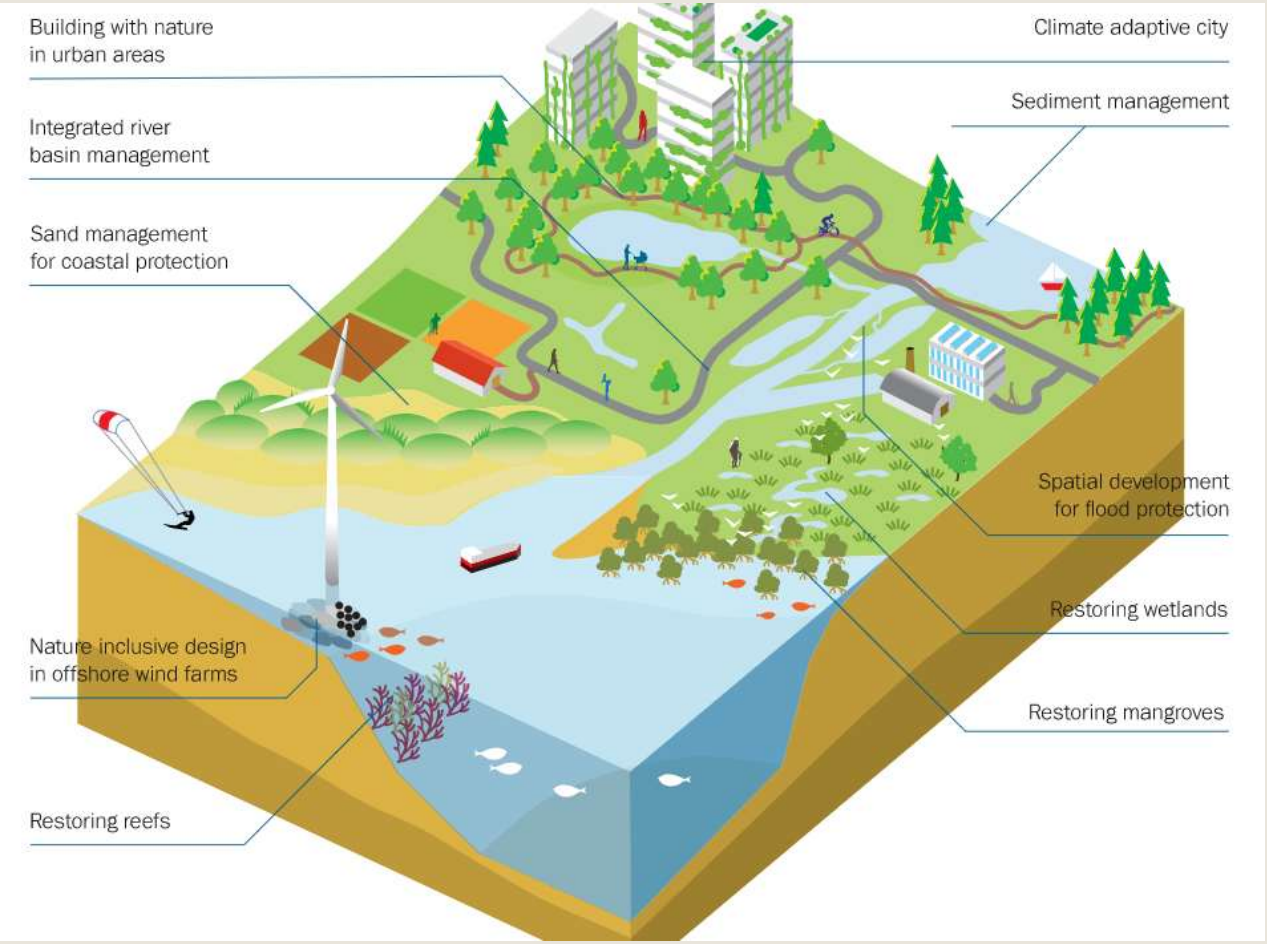
Nature-based solutions (NBS)

- Carson's thoughts in 1962 in *Silent Spring*
  - Alternatives:
    - the selective spraying, as argued *"The success of selective spraying … aimed not at destroying a particular species but at managing vegetation as a living community."*
    - biological control (*e.g. "control a plant with a plant-eating insect"*)

- Modern environmental governance in 21st Century
  - Nature-based solutions as opposed to technical-based solutions to restore nature.

Nature-based solutions (NBS), coastal management as example

NBS refers to the sustainable management and use of nature for tackling socio-environmental challenges.



- For coastal areas, erosion, flooding, and storm surges are primary concerns.
- NBS include sand nourishment, vegetation rehabilitation, wetland and oyster reef restoration…
- NBS are cost effective as well.
- NBS becomes more attractive as climate change and coastal development intensifies in future.

Summary

- The overriding theme of *Silent Spring* is the powerful—and often negative—effect humans have on the natural world.
- Carson's main argument is that pesticides have detrimental effects on the environment; she says these are more properly termed "biocides" because their effects are rarely limited to the target pests.
- DDT is a prime example, but other synthetic pesticides—many of which are subject to bioaccumulation—are scrutinized.
- Carson accuses the chemical industry of intentionally spreading disinformation (虛假信息) and public officials of accepting industry claims uncritically.

Silent Spring – the storm of criticism

- Met with savage and relentless attack by the pesticides industry.

- Rachel Carson was discounted as a "hysterical (歇斯底里) woman," and there were various attempts to discredit both her and her findings.
  - Pesticide producers tried to intimidate Carson's publisher into suppressing the book before publication.
  - Agricultural and trade journals attacked *Silent Spring* before it hit the shelves.
  - Chemical companies threatened to pull ads from magazines and newspapers that gave *Silent Spring* favorable reviews.

Silent Spring – the storm of criticism

- "If man were to follow the teachings of Miss Carson, we would return to the Dark Ages, and the insects and diseases and vermin would once again inherit the earth."
  - Dr. Robert White-Stevens, biochemist, a spokesman for chemical industry
- A one-sided case
  - Quotes only isolated cases of bad effects of one or two pesticides, or pesticide misuse
- Exaggerating the dangers of DDT
  - Repeated studies showed no evidence that DDT increases cancer risk

Silent Spring – Impact

- President John F. Kennedy read it and issued an investigation into pesticides by the President's Science Advisory Committee
- Campaign to ban DDT in the United States
- Environmental movement
- Establishment of the U.S. Environmental Protection Agency in 1970 and the passing of numerous environmental laws (*e.g.* the Clean Air Act in US).
- Ban on use of DDT in 1972 in the US
- Presidential Medal of Freedom (1980)
  - an award bestowed by the president of the U.S. to recognize people who have made "an especially meritorious contribution to the security or national interests of the U.S., world peace, cultural or other significant public or private endeavors".

Carson’s Legacy

- Concept of ecosystem balance
- Question authority and policies
- Public awareness to environmental issues

- Similar examples in 21st Century to human perturbation to the environment?

Plastics nowadays as Pesticides?





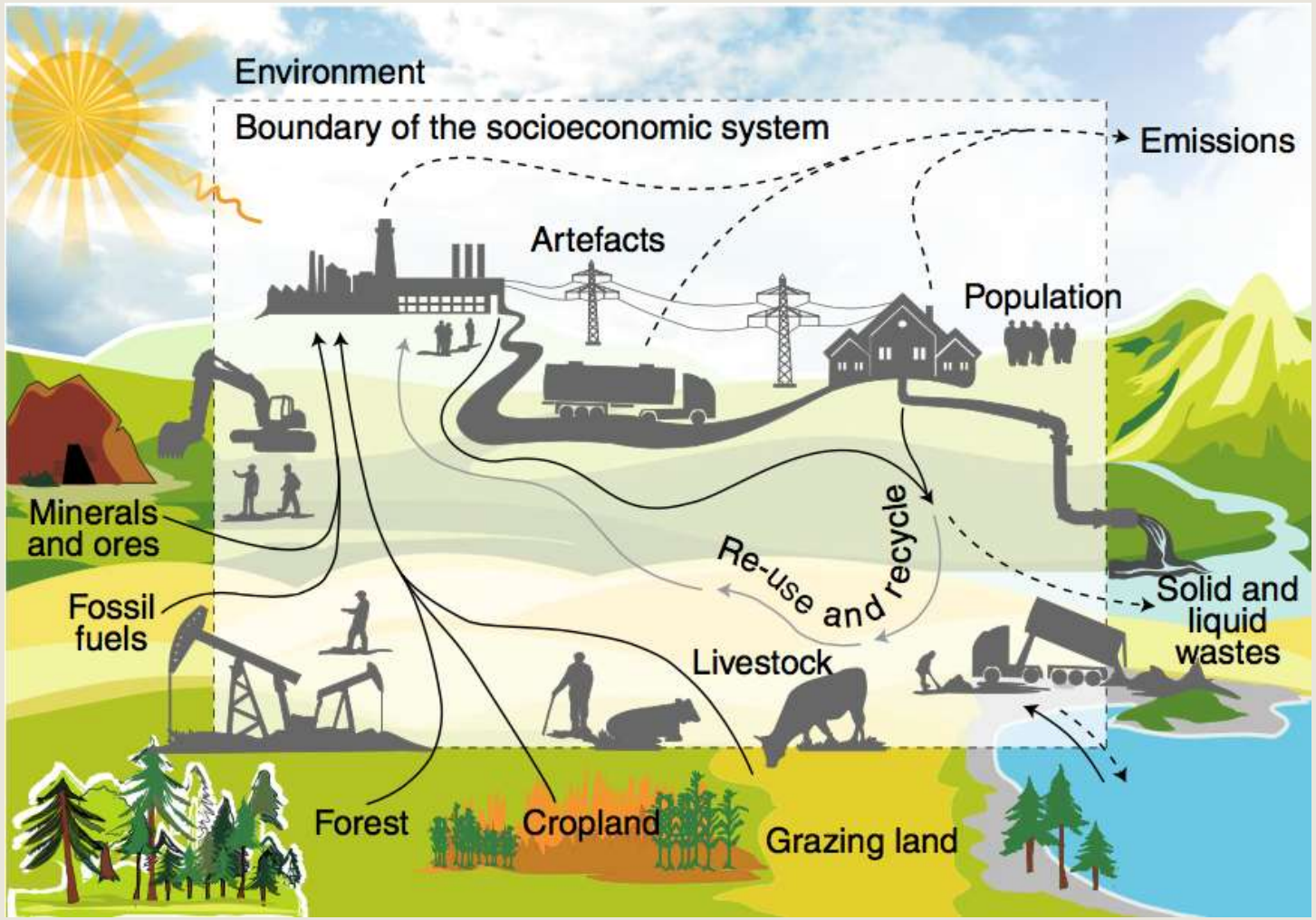
- *GHG (greenhouse gas) emissions from plastics would reach 15% of the global carbon budget by 2050.*
- *Use of bio-based plastics, aggressive application of renewable energy, recycling and demand-management strategies, in concert, has the potential to keep 2050 emissions comparable to 2015 levels.*

The earth's vegetation is part of a web of life in which there are intimate and 2
essential relations between plants and the earth, between plants and other plants, between plants and animals. Sometimes we have no choice but to disturb these relationships, but we should do so thoughtfully, with full awareness that what we do may have consequences remote in time and place. But no such humility marks

- How can we be sure of all the consequences/impacts of our action?
- If we can't predict precisely what will happen, what shall guide our actions and decisions?

Eco-centric mindset

Human rely on nature and the power of healthy ecosystems



- Environmental pressure
  Raw material extraction...
- Environmental impact
  Emisions, wastes...

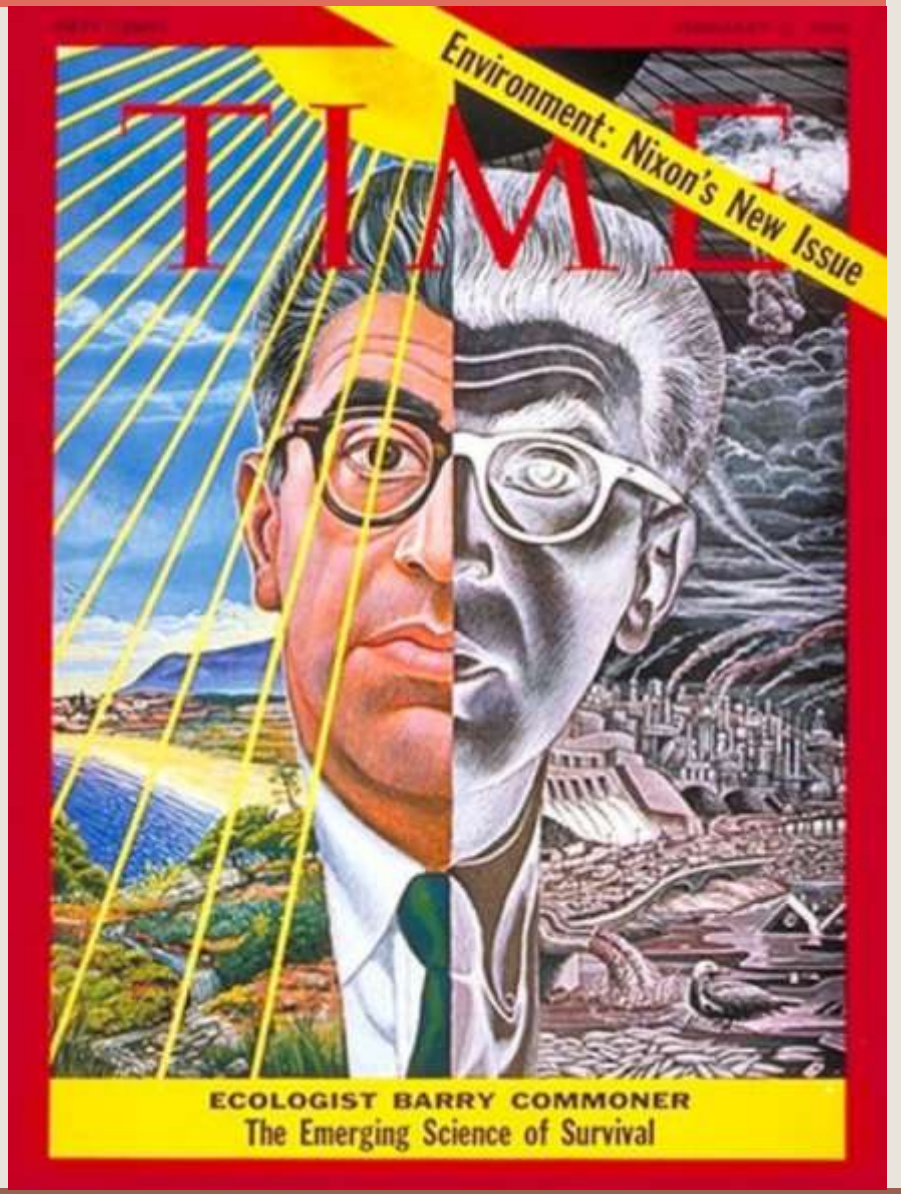
Reflections from Carson's environmental philosophies: Interdependence of Life

- Barry Commoner, a leader among a generation of "scientist-activists" and possibly "the greatest environmentalist of the 20th century".

- In 1970, Time featured Commoner on its cover.

- Known for his four "laws of ecology":

1) Everything is connected to everything else;
2) Everything must go somewhere;
3) Nature knows best;
4) There is no such thing as a free lunch.



Interdependence of Life to eco-centric mindset



The eco-centric ideology, the most mutually beneficial, is to consider yourself as equal value to plants, animals and land.

Eco-centric leadership

- ✓ Each part of earth is inseparable and maintains intrinsic value no matter its monetary value for human use.

Interdependence of Life to eco-centric mindset



An egocentric mindset allows us as individuals to prosper without considering the cost to others.

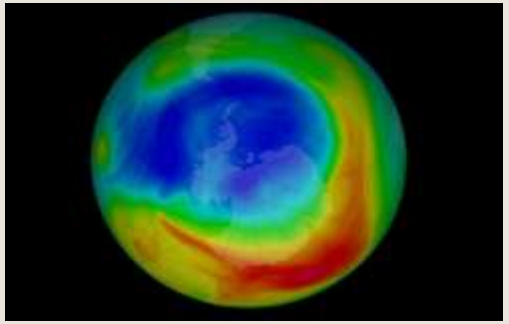
Egocentric leadership is at the root of our environmental crisis.

- ✓ Unfortunately, believing that everyone working on themselves will collectively create good often disregards the cost to non-human organisms and inanimate systems of the biosphere.
- ✓ This narrowing perspective aligns with capitalism, consumerism.

Global Environmental Issue



Climate change



Ozone layer depletion



Forest deterioration



Acid rain



Biodiversity decrease



Land desertification

Environmental Issue Development

- The United Nations Conference on the human environment, June 5-16, 1972, Stockholm, Sweden.
  - The meeting agreed upon a Declaration containing 26 principles concerning the environment and development; an Action Plan with 109 recommendations, and a Resolution.
    《联合国人类环境会议宣言》《人类环境宣言》

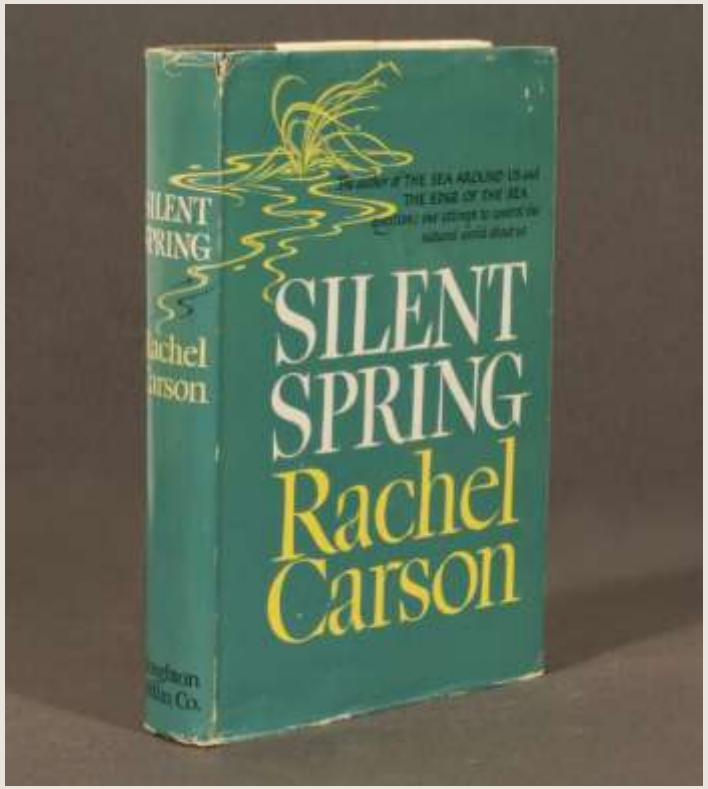
Environmental Issue Development

- *Kyoto Protocol* (1997)
  - aimed to reduce the emission of gases that contribute to global warming

- *Paris Agreement* (2015)
  - legally binding international treaty on climate change






SILENT
SPRING
Rachel
Carson

Foreword

IN 1958, when Rachel Carson undertook to write the book that became *Silent Spring*, she was fifty years old. She had spent most of her professional life as a marine biologist and writer with the U.S. Fish and Wildlife Service. But now she was a world-famous author, thanks to the fabulous success of *The Sea Around Us*, published seven years before. Royalties from this book and its successor, *The Edge of the Sea*, had enabled her to devote full time to her own writing. To most authors this would seem like an ideal situation: an established reputation, freedom to choose one's own subject, publishers more than ready to contract for anything one wrote. It might have been assumed that her next book would be in a field that offered the same opportunities, the same joy in research, as did its predecessors. Indeed she had such projects in mind. But it was not to be. While working for the government, she and her scientific colleagues had become alarmed by the widespread use of DDT and other long-lasting poisons in so-called agricultural control programs. Immediately after the war, when these dangers had already been recognized, she had tried in vain to interest some magazine in an article on the subject. A decade later, when the spraying of pesticides and herbicides (some of them many times as toxic as DDT) was causing wholesale destruction of wildlife and its habitat, and clearly endangering human life, she decided she had to speak out. Again she tried to interest the magazines in an article. Though by now she was a well-known writer, the magazine publishers, fearing to lose advertising, turned her down. For example, a manufacturer of canned baby food claimed that such an article would cause "unwarranted fear" to mothers who used his product. (The one exception was *The New Yorker*, which would later serialize parts of *Silent Spring* in advance of book publication.) So the only answer was to write a book—book publishers being free of advertising pressure. Miss Carson tried to find someone else to write it, but at last she decided that if it were to be done, she would have to do it herself. Many of her strongest admirers questioned whether she could write a salable book on such a dreary subject. She shared their doubts, but she went ahead because she had to. "There would be no peace for me," she wrote to a friend, "if I kept silent."

Silent Spring was over four years in the making. It required a very different kind of research from her previous books. She could no longer recount the delights of the laboratories at Woods Hole or of the marine rock pools at low tide. Joy in the subject itself had to be replaced by a sense of almost religious dedication. And extraordinary courage: during the final years she was plagued with what she termed "a whole catalogue of illnesses."

Also she knew very well that she would be attacked by the chemical industry. It was not simply that she was opposing indiscriminate use of poisons but—more fundamentally—that she had made clear the basic irresponsibility of an industrialized, technological society toward the natural world. When the attack did come, it was probably as bitter and unscrupulous as anything of the sort since the publication of Charles Darwin's *Origin of Species* a century before.

Hundreds of thousands of dollars were spent by the chemical industry in an attempt to discredit the book and to malign the author—she was described as an ignorant and hysterical woman who wanted to turn the earth over to the insects. These attacks fortunately backfired by creating more publicity than the publisher possibly could have afforded. A major chemical company tried to stop publication on the grounds that Miss Carson had made a misstatement about one of their products. She hadn't, and publication proceeded on schedule.

She herself was singularly unmoved by all this furor.

Meanwhile, as a direct result of the message in *Silent Spring*, President Kennedy set up a special panel of his Science Advisory Committee to study the problem of pesticides. The panel's report, when it appeared some months later, was a complete vindication of her thesis. Rachel Carson was very modest about her accomplishment. As she wrote to a close friend when the manuscript was nearing completion: "The beauty of the living world I was trying to save has always been uppermost in my mind—that, and anger at the senseless, brutish things that were being done.... Now I can believe I have at least helped a little." In fact, her book helped to make ecology, which was an unfamiliar word in those days, one of the great popular causes of our time. It led to environmental legislation at every level of government.

Twenty-five years after its original publication, Silent Spring has more than a historical interest. Such a book bridges the gulf between what C. P. Snow called "the two cultures." Rachel Carson was a realistic, well-trained scientist who possessed the insight and sensitivity of a poet. She had an emotional response to nature for which she did not apologize. The more she learned, the greater grew what she termed "the sense of wonder." So she succeeded in making a book about death a celebration of life. Rereading her book today, one is aware that its implications are far broader than the immediate crisis with which it dealt. By awaking us to a specific danger—the poisoning of the earth with chemicals—she has helped us to recognize many other ways (some little known in her time) in which mankind is degrading the quality of life on our planet.

And Silent Spring will continue to remind us that in our over-organized and over-mechanized age, individual initiative and courage still count: change can be brought about, not through incitement to war or violent revolution, but rather by altering the direction of our thinking about the world we live in.

Read the foreword and write a paragraph of reflection on the part that resonated with you the most.

Reading Assignment



Finish reading Text 6.